

Omninet AI: The Internet Everywhere

Marketing Requirements Document – Version 1.3

Ver.	Date	Author	Description	Approved by	Approval Date
1.0	Feb 24, 2026	AI Update, Rex Black	Modernized for AI & Cloud Architecture		
1.1	Mar 24, 2026	AI Update	Incorporated formal inspection findings		
1.2	Mar 26, 2026	AI Update	Addressed follow-up inspection issues		
1.3	Apr 07, 2026	AI Update	Updated based on quality risk analysis		

1. Scope

This document specifies the requirements for a next-generation family of Internet kiosks called **Omninet AI**. These kiosks shall provide users with secure, high-speed, and intelligent Internet access in public places. The system will support modern payment

methods and leverage artificial intelligence to enhance user experience, security, and operational efficiency.

1.1 Terms, Acronyms, and Abbreviations

(No changes – same as v1.2)

1.2 Applicable Documents

(No changes – same as v1.2)

2. Required Release Date

The first set of 1,000 Omninet AI kiosks shall be live, accepting payment and accessing the Internet, by September 30, 2024.

3. Description of Requirements

3.1 General Technical Requirements

Omninet AI shall provide users in airports, malls, hotels, and other public places with secure, personalized, and high-speed Internet access.

The Omninet AI system comprises three primary subsystems:

1. The Kiosk (Edge Device)
2. The Omninet Cloud (Core Services)
3. The AI Support Center

[[NEW]] The cloud infrastructure shall support at least 10,000 concurrently active kiosks and scale elastically.

[[NEW]] All communication between kiosk and cloud shall be encrypted using TLS 1.3 (or higher).

3.1.1 Welcome

(No change – same as v1.2)

3.1.2 Payment

Once a user initiates a session, the kiosk shall offer flexible payment options. The system shall use AI to suggest optimal session lengths based on location and historical usage patterns; details of AI model training and updates are defined in the System Requirements Document [6] and the AI Governance Document [4].

The system accepts the following forms of payment:

- Traditional: Bills and coins (see screen prototype K.3). For cash, the kiosk shall accept denominations up to \$20 and shall return change.
- Modern Cards: Credit/Debit cards with chip and NFC support.
- Digital & Mobile: Mobile wallets (Apple Pay, Google Pay), and QR code payments (e.g., WeChat Pay, Venmo).
- AI-Powered Promotions: The system may offer promotional time in exchange for viewing an ad or completing a quick survey.

Time is sold in flexible increments, with a minimum of 1 minute and a maximum of 2 hours. An AI agent will predictively prompt the user to purchase more time based on their browsing activity and remaining time (see screen prototype K.9). The pop-up message shall offer a 'Yes' button that takes the user to the payment screen, and a 'No' button that dismisses the message; the message shall appear every 30 seconds until the session ends or the user acts.

[[NEW]] Payment authorization (excluding user input time) shall complete within 5 seconds under normal network conditions. If authorization exceeds 20 seconds, the kiosk shall present a timeout message and allow the user to retry or cancel.

3.1.3 Internet Browser & AI Assistant

At the Welcome screen, each Omninet AI kiosk shall launch a secure, modern browser supporting the latest versions of Chromium, Firefox, and Safari. The browser will be integrated with an **Omninet AI Assistant**—a conversational AI agent that can:

- Answer questions about the kiosk's features.
- Help find information or navigate to websites.
- Translate on-screen text or web pages in real-time.
- Summarize articles or content for the user.

[[NEW]] The AI Assistant shall respond to user queries within 2 seconds (95th percentile) under normal network and server load.

3.1.4 Performance

All kiosks shall have a minimum connection speed of 100 MBPS, provided via 5G, WiFi6, or Fiber backhaul.

[[NEW]] After a session terminates, the kiosk shall return to the Welcome screen within 10 seconds (95th percentile) and within 20 seconds maximum.

3.1.5 Localization

Each Omninet AI kiosk shall detect the most likely user language based on location or offer a selection at the Welcome screen. **[[REVISED]] The language selection shall be presented as a list of language names in their native script (e.g., “中文” for Chinese) and also in the current selected language, to avoid confusion.** The language selection shall persist for the entire session and be stored in local session data; the system shall support at least 20 languages including English, Spanish, French, Mandarin, Arabic, etc. The integrated AI Assistant shall support real-time conversational translation, allowing users to interact in their preferred language for at least 20 languages.

The browser shall support Unicode (UTF-8) and render any language that uses Unicode characters; the AI Assistant shall provide real-time translation for at least 20 languages.

3.1.6 Content Control & Security

Omninet AI shall employ a multi-layered security approach:

- AI-Powered Content Filtering: Omninet shall implement an AI-based content filter that blocks or blurs material defined as ‘adult’ or ‘violent’ according to the Omninet Content Filtering Policy (document TBD). **[[NEW]] The filter shall be localised per region to respect local cultural and legal norms.**
- Advanced Threat Protection: Omninet shall employ an on-device AI security agent that scans incoming/outgoing traffic using updated threat definitions; detection shall result in session termination and a security alert.

[[NEW]] The AI Security Agent shall be updated with new threat definitions at least once every 24 hours, and critical updates may be pushed immediately.

3.1.7 Session Termination

Users may terminate sessions by:

- Logging out (no refund is given for unused time).
- Allowing purchased time to expire (no refund).

If a call center agent terminates a session for inappropriate activity (see section 3.2.5), the user shall receive a refund for any unused time.

[[NEW]] The logout function shall be accessible from the browser taskbar at all times during an active session.

3.1.8 Confidentiality & Privacy

To ensure complete user privacy, upon session termination, each kiosk shall:

- Execute a secure wipe of all session data (cache, cookies, history, downloaded files) from the local device.
- Restart the browser in a fresh, sandboxed environment at the Welcome screen.
- All stored session data, including AI Assistant conversation logs, shall be securely deleted from local storage; any cloud backups shall be deleted within 24 hours.

[[NEW]] The secure wipe shall use methods that prevent data recovery from solid-state storage (e.g., TRIM command or cryptographic erasure).

3.2 Administration (via Omninet Cloud & AI Support Center)

3.2.1 Software & AI Model Updates

Software and security updates shall be managed centrally from the Omninet Cloud.

- At 2:00 AM local time, each kiosk shall check for and apply updates, including:
 - OS and browser patches.
 - Device drivers.
 - AI Agent behavioural models and content filtering datasets.
 - Updated payment rate tables.
 - Dynamic threat intelligence feeds.
- The update process is managed by an **AI Orchestrator Agent** in the cloud. If the update server returns a 503 (Service Unavailable) or similar overload response, the kiosk shall retry after a randomised delay between 10 and 60 minutes. The retry count is capped at 3 attempts; after that, the kiosk shall log an error and alert the support center.
- Critical security updates or AI model patches can be pushed immediately by the AI Support Center.

[[NEW]] Updates shall be designed to minimise bandwidth impact during peak kiosk usage hours; large downloads (e.g., AI models) may be throttled or delivered via peer-to-peer between kiosks where permitted.

3.2.2 View Kiosks (AI-Powered Monitoring)

(No change – same as v1.2)

3.2.3 View Users (with AI Oversight)

(No change – same as v1.2)

3.2.4 Modify User

(No change – same as v1.2)

3.2.5 Terminate User & AI-Enhanced Intervention

(No change – same as v1.2)

Appendix A: Referenced Documents

(No change – same as v1.2)

Summary of Changes in v1.3

Section	Change Type	Description
3.1 (new)	Addition	Added requirement for cloud scalability (10,000 kiosks).
3.1 (new)	Addition	Added encryption requirement (TLS 1.3).
3.1.2	Addition	Added payment processing time limit (≤ 5 seconds) and timeout handling.
3.1.3	Addition	Added AI Assistant response time requirement (≤ 2 seconds).
3.1.4	Addition	Added welcome screen reload time requirement (≤ 10 seconds).
3.1.5	Revision	Clarified language selection presentation (native script + current language).

Section	Change Type	Description
3.1.6	Addition	Added requirement for localised content filtering and AI Security Agent update frequency.
3.1.7	Addition	Added requirement that logout is always accessible.
3.1.8	Addition	Added detail on secure wipe for SSDs.
3.2.1	Addition	Added bandwidth management for updates.

All changes are marked with **[[NEW]]** or **[[REVISED]]** in the document body. These updates address the highest-priority risks identified in the quality risk analysis (e.g., performance, reliability, security, usability) while preserving the original scope and intent of the Marketing Requirements Document.